

TEZPUR UNIVERSITY
End-Semester Examination, Autumn 2015
CO 504 / CS 537 NATURAL LANGUAGE
PROCESSING

Marks : 60

Time : 2 hours

The figures in the right-hand margin indicate marks for the individual question.

Answers should be concise and entire answer to a question should be together. Give examples and diagrams wherever appropriate. State assumptions wherever made.

Submit question no. 14 within first fifteen minutes.

1. What is *IPA* in speech encoding? What are its advantages in comparison to the conventionally used scripts? 2 + 2
2. What is *morphology* in linguistics? How does morphology influence word recognition in natural language processing? 2 + 2
3. How is *n-gram* based text analysis sensitive to the training corpus? What is a *balanced corpus*? 2 + 2
4. (a) With an example, explain why part-of-speech tagging is not a trivial exercise? 3
(b) Perform PoS tagging for the following sentence- 3
I want to purchase a gift for him from the market.
5. Give examples of different forms of noun phrases and write context free grammar rules for the different forms. 2 + 3
6. How is Earley's parsing technique more suitable for parsing natural languages in comparison to the parsing techniques for programming languages? 2

7. What aspect of syntax analysis is covered by *unification of feature structures*? 3

8. Why is a TreeBank required to implement *probabilistic context free grammar*? 2

9. Represent the following sentence using First Order Predicate Calculus- 2

Ram will telephone Sita from his office.

10. Identify lexical semantic relationships among the following words- 3

table (furniture), have (own), current (electricity), eat, table (information format in documents), current (flow of river), it, furniture, have (consume food).

11. Give an example to illustrate the problem of reference resolution in a discourse. How can gender and number help in resolving references. 2 + 2

12. How are dialogue and monologue discourses different? 2

13. Why is *deeper semantic analysis* required in *interlingua* based machine translation than in *transfer model* of translation? 3

Natural Language Processing

Roll No. _____

Detach this page and submit it within first fifteen minutes.

14. (I) Fill in the blanks 1 × 8

(a) Nyquist-Shannon sampling theorem is relevant in _____ (speech / text) encoding.

(b) The regular expression $/^{\wedge} [Tt]he[rs]e/$ would _____ (accept / reject) the string *these*.

(c) Obtaining the word *government* from the word *govern* is an instance of _____ (derivation / inflection).

(d) The proportion of relevant documents retrieved out of all documents retrieved is _____ (recall / precision).

(e) “Before Ram bought the coat, he compared the prices in several shops”. In this sentence, *he* is a _____ (cataphora/anaphora).

(f) *Grounding* is a phenomenon observed in _____ (monologue / dialogue).

(g) When translation is required between one another among several languages _____ (direct / interlingua based) translation is more appropriate approach.

(h) The *fluency* of sentences produced via machine translation, can be assessed using _____ (n-grams/parallel corpora).

(II) State whether each of the following statements are true or false- 1×8

- (a) *Chair* is a synonym of *furniture*.
- (b) In the sentence “Ram has a truck” *has* is the main verb.
- (c) The sentences “Ram came to office this morning. Kamal does not like sweets.” lack coherence.
- (d) In Unicode BMP is one of several planes of 64 K codes.
- (e) Polysemy may lower the precision of information retrieval.
- (f) In the sentence “Kamal said that Vimal bought himself a new shirt” *himself* refers to Kamal.
- (g) The higher the entropy of a system, the higher is the predictability of its behaviour.
- (h) Machine translation is more effective in restricted domains.